



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO AS PETROLEUM

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

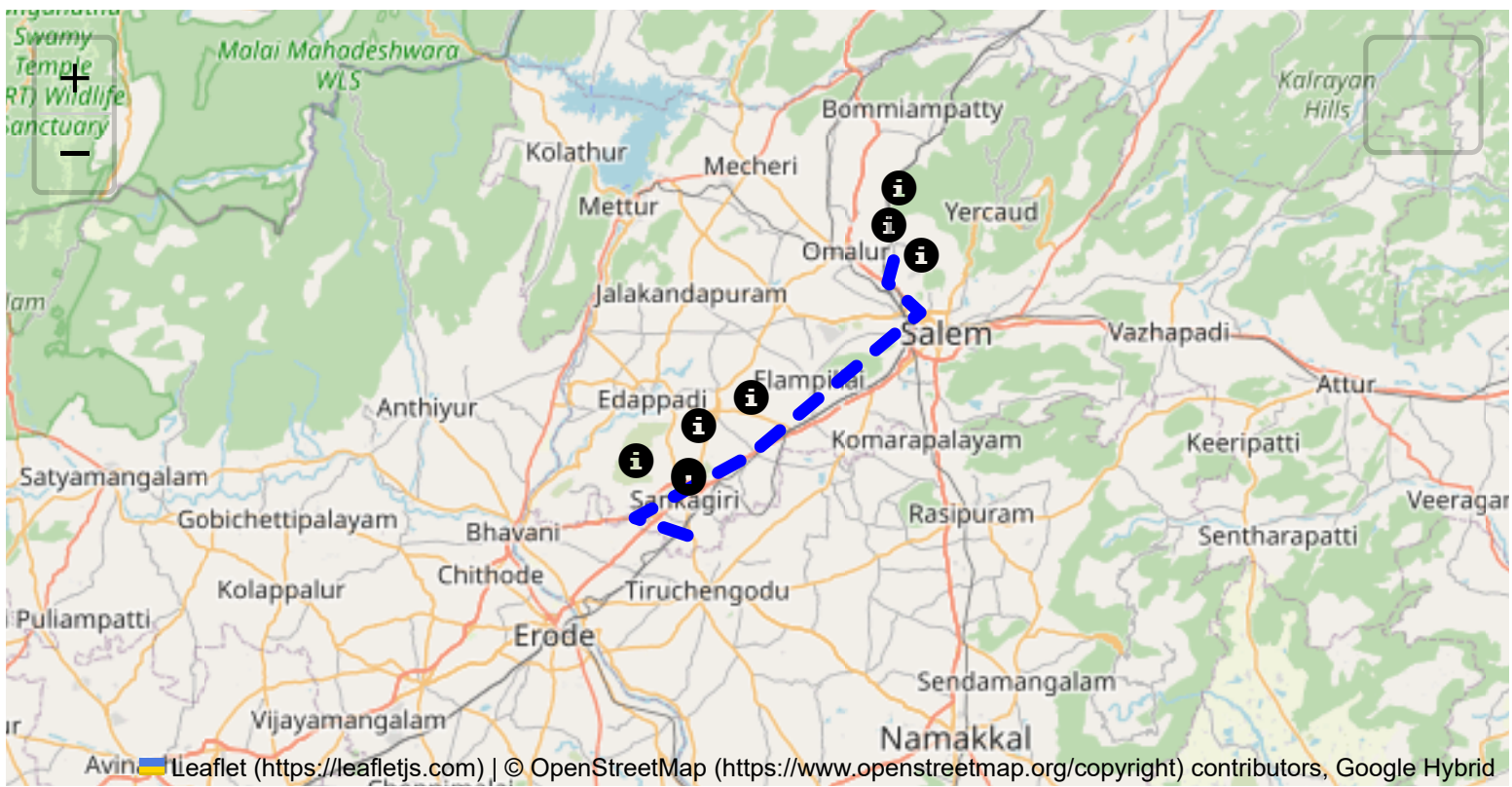
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

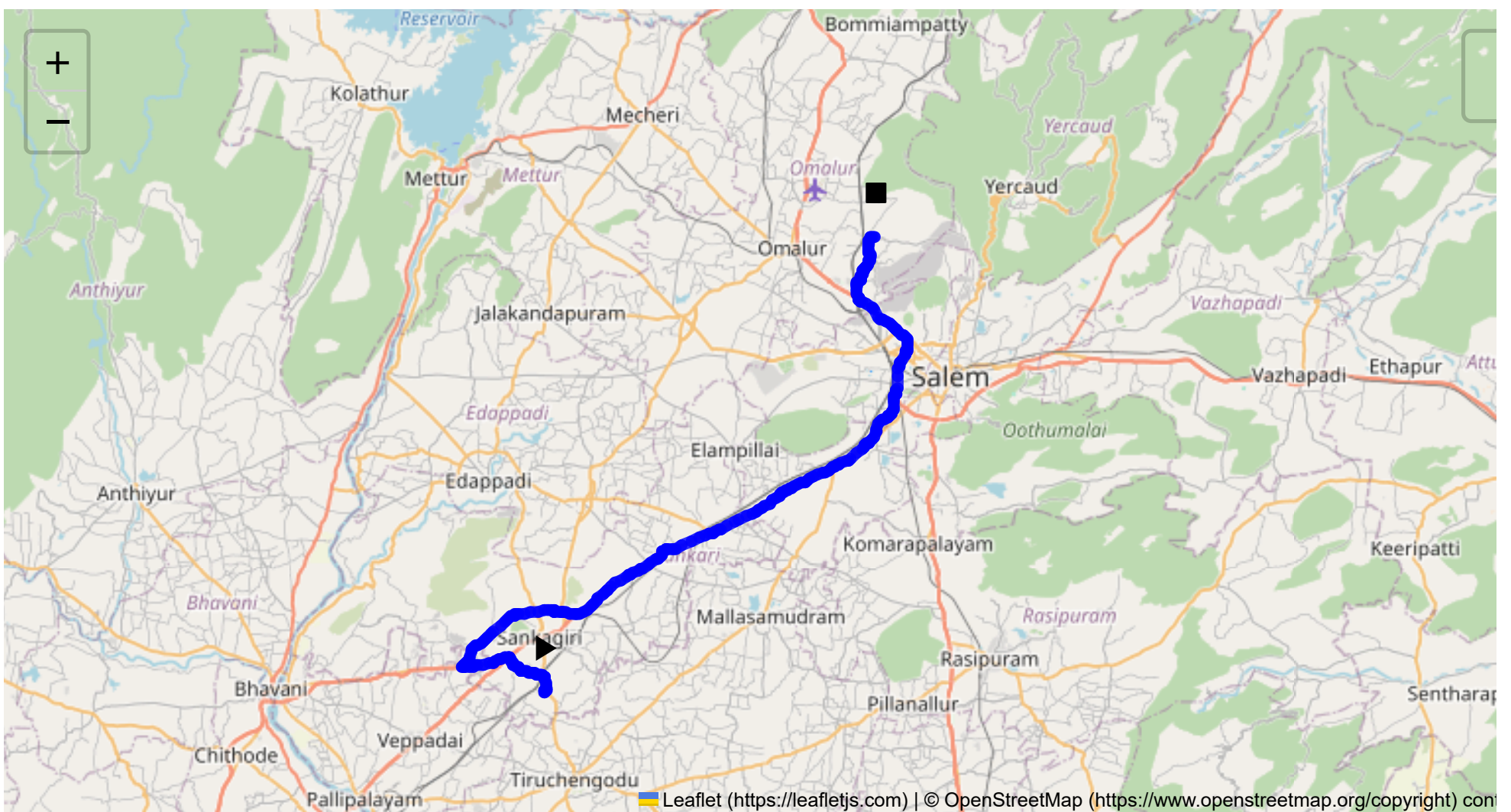
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 64.75 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.8 hours
Start: (11.4381, 77.8734)
End: (11.7497615, 78.1048036)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

The route begins at CVQF+23W in Sangagiri, passing through Puthur, Padaiveedu, and areas along Salem-Kochi Highway, before reaching the destination in Vellalapatti. The journey covers a range of rural and semi-urban roads, including state highways and bypass roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- **Climate:** Generally, Tamil Nadu experiences a tropical climate with three major seasons: summer (March to May), monsoon (June to December), and winter (January to February).
- **Hazards:** During monsoon, heavy rainfall can lead to poor visibility, waterlogging, and slippery roads. Summer can cause road surface expansion and occasionally severe heat-related vehicle issues.

3. Traffic Patterns:

- **Peak Hours:** Early morning (7-9 AM) and late evening (5-7 PM) are typical congestion times.
- **Congestion-Prone Areas:** Salem's Bangalore bypass road and Karuppur Railway Flyover Bridge are known for heavy traffic.

4. Assessment of Road Quality and Infrastructure:

- **Road Condition:** Overall, road quality is moderate, with highways generally well-maintained, but rural roads might have potholes and uneven surfaces.
- **Infrastructure:** Adequate for regular traffic; however, improvements might be necessary for managing heavy vehicle and hazardous materials transit.

5. Suggestions for Alternative Routes for Emergencies:

- Consider using Salem Ring Road via NH 44 for a detour if roads are blocked due to weather or accidents.
- Lesser-known village roads might be available but require slower speeds and caution due to poor quality.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Strict compliance with Tamil Nadu's regulations for transport of hazardous materials is mandatory.
- Drivers should possess proper certification, and vehicles must be equipped with safety kits and necessary labels.

7. Overview of Historical Incidents:

- Accidents involving heavy vehicles have occurred, particularly during monsoon due to poor visibility and slick roads.
- Incidents in congested areas around Salem were often related to brake failures and overloading.

8. Environmental Considerations and Sensitive Areas:

- **Protected Areas:** Ensure caution near any wildlife reserves or ecologically sensitive zones.
- **Pollution:** Monitor for emission limits in urban areas to avoid fines and ensure cleaner transport.

9. Analysis of Communication Coverage:

- Generally good coverage in urban and semi-urban areas.
- Potential dead zones include rural stretches, especially near Padaiveedu and Vaikundam. It is advisable to have communication alternatives.

10. Estimated Emergency Response Times:

- **Urban Areas (Salem vicinity):** Around 30-45 minutes due to better infrastructure and availability.
- **Rural Areas:** 60-90 minutes due to longer distances and limited facilities.

11. Overall Summary of Risk Assessment:

This route presents moderate risk primarily due to weather influences, road conditions, and traffic congestion. Adequate preparations, understanding of peak traffic times, and adherence to local transportation regulations can mitigate most risks. Awareness of historical incidents and preparation for communication gaps will also improve safety. Alternative routes and contingency plans must be in place to handle emergencies effectively.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43961, 77.87341	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44898, 77.87410	30 KM/Hr
4	Turn	Medium	11.45357, 77.85789	30 KM/Hr
5	Turn	Medium	11.45352, 77.85698	30 KM/Hr
6	Turn	Medium	11.45847, 77.85140	30 KM/Hr
7	Turn	Medium	11.46303, 77.84945	30 KM/Hr
8	Turn	Medium	11.46314, 77.84928	30 KM/Hr
9	Turn	Medium	11.45543, 77.81740	30 KM/Hr
10	Turn	Medium	11.45563, 77.81708	30 KM/Hr
11	Turn	High	11.45631, 77.81668	15 KM/Hr
12	Turn	High	11.49409, 77.88004	15 KM/Hr
13	Turn	Medium	11.49419, 77.88015	30 KM/Hr
14	Turn	Medium	11.49365, 77.88596	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Turn	High	11.49236, 77.89552	15 KM/Hr
16	Turn	High	11.49225, 77.89560	15 KM/Hr
17	Turn	Medium	11.60942, 78.10056	30 KM/Hr
18	Turn	Medium	11.60965, 78.10070	30 KM/Hr
19	Turn	Medium	11.64310, 78.11982	30 KM/Hr
20	Turn	High	11.69388, 78.10965	15 KM/Hr
21	Turn	High	11.69400, 78.10963	15 KM/Hr
22	Turn	Medium	11.70757, 78.09323	30 KM/Hr
23	Turn	Medium	11.70821, 78.09318	30 KM/Hr
24	Turn	High	11.70828, 78.09323	15 KM/Hr
25	Turn	Medium	11.70855, 78.09304	30 KM/Hr
26	Turn	Medium	11.71616, 78.09287	30 KM/Hr
27	Turn	High	11.71690, 78.09240	15 KM/Hr
28	Turn	High	11.71755, 78.09256	15 KM/Hr
29	Turn	Medium	11.71816, 78.09185	30 KM/Hr
30	Turn	Medium	11.71829, 78.09183	30 KM/Hr
31	Turn	High	11.72096, 78.09477	15 KM/Hr
32	Turn	Medium	11.72215, 78.09520	30 KM/Hr
33	Turn	High	11.72240, 78.09501	15 KM/Hr
34	Turn	Medium	11.73016, 78.09877	30 KM/Hr
35	Turn	High	11.73046, 78.09949	15 KM/Hr
36	Turn	High	11.73670, 78.10229	15 KM/Hr
37	Turn	Medium	11.74803, 78.10069	30 KM/Hr
38	Turn	Medium	11.74967, 78.10232	30 KM/Hr
39	Turn	Medium	11.74996, 78.10236	30 KM/Hr
40	Turn	High	11.75058, 78.10279	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
14	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
15	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
16	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
17	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
19	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
21	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
22	hospital	Global Medical Centre Jagirammalayam	11.6894409, 78.1108847	30 km/h	Medium
23	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
24	hospital	Rajaam Hospital	11.7212623, 78.0951467	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
18	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
20	school	Mothers Vidhyalaya Metric School	11.6884655, 78.1240569	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45563, 77.81708



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49409, 77.88004



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88015



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49365, 77.88596



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49236, 77.89552



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49225, 77.89560



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60942, 78.10056



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60965, 78.10070



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64310, 78.11982



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69388, 78.10965



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69400, 78.10963



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70757, 78.09323



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70821, 78.09318



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70828, 78.09323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70855, 78.09304



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71616, 78.09287



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.71690, 78.09240



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.71755, 78.09256



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71816, 78.09185



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71829, 78.09183



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.72096, 78.09477



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72215, 78.09520



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.72240, 78.09501



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73016, 78.09877



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73046, 78.09949



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73670, 78.10229



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74803, 78.10069



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74967, 78.10232



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74996, 78.10236



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.75058, 78.10279
